Mathematical Methods for Finance

Module 2 – L^p Spaces, Fourier Methods and Approximation

Lesson 8: Spaces of functions and frequency tools

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3-hour lesson structure: intuition, theory, examples, exercises, and exam-style practice

Learning outcomes for Lesson 8

By the end of this lesson, students should be able to:

1. define L^p spaces and compute basic L^p norms;
2. use Hölder and Minkowski in integral form;
3. explain why L^2 is a Hilbert space;
4. compute elementary Fourier coefficients;
5. describe Fourier transform, convolution, Plancherel and FFT at an intuitive level.

Exam relevance

This lesson provides the language needed to read the next modules: measurable sets, measurable functions and measures are the base layer of integration and probability.

How today is organised

14:30–15:20

L^p spaces
Norms, equivalence classes and inequalities.

15:35–16:25

L^2 and Fourier series
Orthogonality, coefficients and approximation.

16:40–17:30

Fourier transform and practice
Convolution, Plancherel, FFT and exam exercises.

Intuition: Learning rule

For each topic: identify the object, write the definition, test it on a small example, then solve one exam-style exercise.

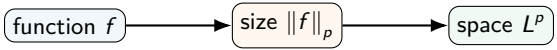
Module 2

Block 1: L^p spaces

From integrable functions to spaces

Intuition: The shift

After defining integrals, we can treat functions as points in a space and measure their size.



Bridge with Module 1

Normed spaces now become spaces of functions, not only finite-dimensional vectors.

L^p spaces: definition

Definition: L^p space

Let $(X, \mathcal{F}, \mu)$ be a measure space and $1 \leq p < \infty$. Define

$$L^p(X) = \left\{ f \text{ measurable} : \int_X |f|^p \, d\mu < \infty \right\},$$

where functions equal almost everywhere are identified. The norm is

$$\|f\|_p = \left(\int_X |f|^p \, d\mu \right)^{1/p}.$$

Definition: L^∞

L^∞ contains essentially bounded functions, with norm

$$\|f\|_\infty = \operatorname{ess\,sup} |f|.$$

Why equivalence classes?

Example: Same L^p object

On $[0,1]$, define

$$f(x) = 0, \qquad g(x) = \begin{cases} 1, & x = 1/2, \\ 0, & x \neq 1/2. \end{cases}$$

Then $f = g$ almost everywhere and

$$\|f - g\|_p = 0.$$

In L^p, they represent the same element.

Interpretation

In integration-based spaces, changing a function on a null set is invisible.

Compute an L^p norm

Worked example: Power function

On $[0,1]$, let $f(x) = x$. Then

$$\|f\|_p = \left(\int_0^1 x^p \, dx \right)^{1/p} = \left(\frac{1}{p+1} \right)^{1/p}.$$

For $p = 2$,

$$\|f\|_2 = \sqrt{\frac{1}{3}}.$$

Exercise: Quick computation **

Compute $\|1\|_p$ on $[0,2]$.

Solution: norm of a constant

Proof idea: Computation

On $[0, 2]$,

$$\|1\|_p = \left(\int_0^2 1^p dx \right)^{1/p} = 2^{1/p}.$$

For $p = \infty$,

$$\|1\|_\infty = 1.$$

Hölder inequality in integral form

Theorem: Hölder

Let $1 < p, q < \infty$ with

$$\frac{1}{p} + \frac{1}{q} = 1.$$

If $f \in L^p$ and $g \in L^q$, then $fg \in L^1$ and

$$\int |fg| \, d\mu \leq \|f\|_p \, \|g\|_q .$$

Special case

For $p = q = 2$, this is Cauchy–Schwarz:

$$\int |fg| \, d\mu \leq \|f\|_2 \, \|g\|_2 .$$

Minkowski inequality in integral form

Theorem: Minkowski

For $1 \leq p < \infty$,

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p .$$

Intuition: Meaning

Minkowski is the triangle inequality in L^p. It is the reason $\|\cdot\|_p$ is a norm.

Exercise: Exam check **

Which property of a norm is guaranteed by Minkowski?

Solution: Minkowski and norms

Proof idea: Answer

Minkowski gives the triangle inequality:

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p .$$

Together with positivity and homogeneity, this makes L^p a normed space after identifying functions equal almost everywhere.

Banach property

Theorem: Completeness of L^p

For $1 \leq p \leq \infty$, the space $L^p(X)$ is complete. Hence L^p is a Banach space.

Intuition: Meaning

If a sequence of functions is Cauchy in L^p, then it converges in L^p to a function that still belongs to L^p.

Connection

This is the function-space version of the completeness principle from Module 1.

Common confusion: pointwise versus L^p

Counterexample: Different convergence concepts

Pointwise convergence and L^p convergence are different.

Example: Spikes again

Let $f_n = n\mathbf{1}_{(0,1/n)}$ on $(0,1)$. Then $f_n \to 0$ pointwise, but

$$\|f_n\|_1 = 1.$$

So f_n does not converge to 0 in L^1.

Module 2

Block 2: L^2 Hilbert space and Fourier series

L^2 as a Hilbert space

Definition: L^2 inner product

For $f, g \in L^2(X)$, define

$$\langle f, g \rangle = \int_X f(x) g(x)\, d\mu(x)$$

in the real case. Then

$$\|f\|_2 = \sqrt{\langle f, f \rangle}.$$

Theorem: Hilbert property

L^2 is a Hilbert space: it is complete and has an inner product.

Orthogonality of functions

Definition: Orthogonality in L^2

Two functions $f, g \in L^2$ are orthogonal if

$$\int f(x)g(x)\,d\mu(x) = 0.$$

Worked example: Example on $[-1, 1]$

Let $f(x) = 1$ and $g(x) = x$. Then

$$\int_{-1}^{1} x\,dx = 0.$$

Therefore 1 and x are orthogonal in $L^2([-1, 1])$.

Trigonometric polynomials

Definition: Trigonometric polynomial

A trigonometric polynomial is a finite linear combination of

$$1, \quad \cos(nx), \quad \sin(nx), \qquad n = 1, 2, \ldots.$$

For example,

$$p(x) = a_0 + a_1 \cos x + b_1 \sin x + a_2 \cos(2x).$$

Intuition: Meaning

Trigonometric polynomials approximate periodic functions using waves.

Orthogonality of sine and cosine

Theorem: Basic orthogonality

On $[-\pi, \pi]$:

$$\int_{-\pi}^{\pi} \cos(nx)\cos(mx)\,dx = 0 \quad (n \neq m),$$

$$\int_{-\pi}^{\pi} \sin(nx)\sin(mx)\,dx = 0 \quad (n \neq m),$$

$$\int_{-\pi}^{\pi} \sin(nx)\cos(mx)\,dx = 0.$$

Meaning

Sines and cosines play the role of perpendicular axes in a function space.

Fourier coefficients

Definition: Fourier coefficients

For a suitable 2π-periodic function f,

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx,$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos(nx)\,dx, \qquad b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin(nx)\,dx.$$

The Fourier series is

$$\frac{a_0}{2} + \sum_{n=1}^{\infty}\big(a_n\cos(nx) + b_n\sin(nx)\big).$$

Worked example: constant function

Worked example: Fourier coefficients of $f(x) = 1$

On $[-\pi, \pi]$:

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} 1 \, dx = 2.$$

For $n \geq 1$,

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} \cos(nx) \, dx = 0,$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} \sin(nx) \, dx = 0.$$

Thus the Fourier series is $a_0/2 = 1$.

Exercise: Fourier coefficients of $\cos x$

Exercise: Coefficient computation ***

Let $f(x) = \cos x$ on $[-\pi, \pi]$. Determine a_0, a_1, b_1, and state what happens to the other coefficients.

Solution: Fourier coefficients of $\cos x$

Proof idea: Use orthogonality

Since $\cos x$ has zero average,

$$a_0 = 0.$$

Also

$$a_1 = \frac{1}{\pi} \int_{-\pi}^{\pi} \cos^2 x \, dx = 1,$$

and

$$b_1 = \frac{1}{\pi} \int_{-\pi}^{\pi} \cos x \sin x \, dx = 0.$$

All other coefficients vanish by orthogonality.

Bessel inequality

Theorem: Bessel inequality

If (e_n) is an orthonormal system in a Hilbert space and f is a vector, then

$$\sum_{n=1}^{\infty} |\langle f, e_n \rangle|^2 \leq \|f\|^2 .$$

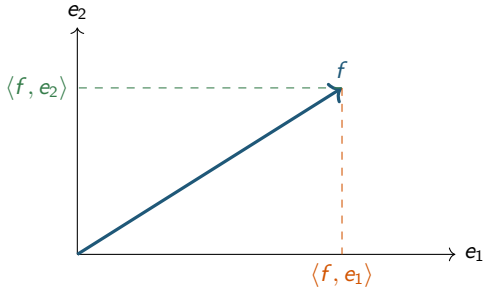
Intuition: Meaning

The total squared size of the coefficients cannot exceed the squared size of the function.

Approximation by projection

Intuition: Fourier as projection

Fourier coefficients are projection coefficients onto orthogonal waves.



Mean-square approximation

Theorem: Projection principle

The best approximation of f in a finite-dimensional subspace is its orthogonal projection onto that subspace.

Example: Fourier partial sum

The N-th Fourier partial sum is the projection of f onto the span of

$$1, \cos x, \sin x, \ldots, \cos(Nx), \sin(Nx).$$

Connection with Module 1

Hilbert projection appears again, now in a concrete function space.

Module 2

Block 3: Fourier transform, convolution and FFT

Fourier transform: idea

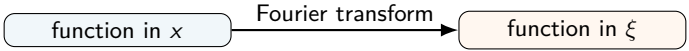
Definition: Fourier transform

For suitable functions $f : \mathbb{R} \to \mathbb{C}$, the Fourier transform is

$$\widehat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{-ix\xi} \, dx.$$

Intuition: Meaning

The Fourier transform decomposes a function into frequencies.



Schwartz space

Definition: Schwartz space

The Schwartz space $\mathcal{S}(\mathbb{R})$ consists of smooth functions that, together with all derivatives, decrease faster than any power at infinity.

Intuition: Why introduced?

Schwartz functions are a clean class where Fourier transforms behave very well.

Example: Gaussian

The function $f(x) = e^{-x^2}$ belongs to $\mathcal{S}(\mathbb{R})$.

Plancherel theorem

Theorem: Plancherel

With the appropriate normalization, the Fourier transform preserves the L^2 norm:

$$\left\|\widehat{f}\right\|_2 = \|f\|_2 .$$

Intuition: Meaning

Energy in the original domain equals energy in the frequency domain.

Hilbert-space message

The Fourier transform is a geometry-preserving transformation on L^2.

Convolution

Definition: Convolution

For suitable functions $f, g : \mathbb{R} \to \mathbb{R}$, define

$$(f * g)(x) = \int_{\mathbb{R}} f(x - y) g(y)\, dy.$$

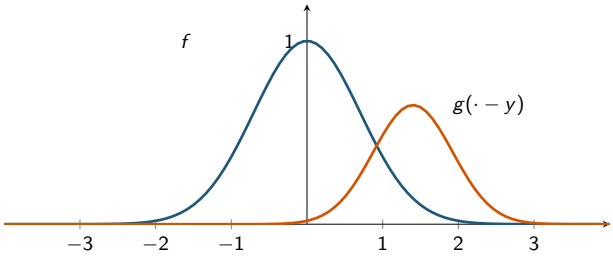
Intuition: Meaning

Convolution blends one function with a shifted version of another.

Example: Smoothing

Convolution with a smooth kernel can turn a rough function into a smoother one.

Picture of convolution



Intuition: Geometric view

Move one function across the other, multiply, integrate the overlap.

Fourier transform of convolution

Theorem: Convolution theorem

For suitable functions,

$$\widehat{f * g} = \widehat{f}\,\widehat{g}.$$

Intuition: Meaning

Convolution in the original domain becomes multiplication in the frequency domain.

Practical consequence

Some complicated integral operations become simpler after applying the Fourier transform.

Fast Fourier Transform

Definition: FFT

The Fast Fourier Transform is an efficient algorithm for computing the discrete Fourier transform.

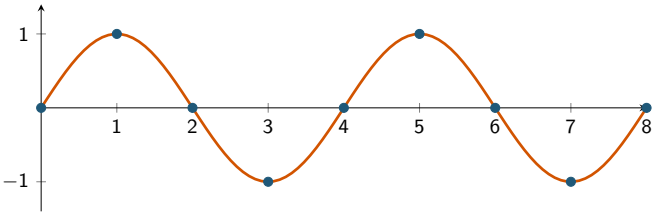
Intuition: Why it matters

The FFT makes frequency computations feasible for large datasets and numerical grids.

Course role

We do not study the algorithm in detail. We need to know what problem it solves and why Fourier methods are computationally useful.

Discrete frequency intuition



Intuition: Message

The discrete Fourier transform decomposes sampled data into oscillating components.

Exercise: identify the correct object

Exercise: Classification **

Classify each expression:

1. $\int_0^1 |f(x)|^2 dx$;
2. $\int_{-\pi}^{\pi} f(x)\cos(nx)dx$;
3. $\int_{\mathbb{R}} f(x-y)g(y)dy$;
4. $\int_{\mathbb{R}} f(x)e^{-ix\xi}dx$.

Solution: identify the correct object

Proof idea: Answers

1. L^2 norm squared.
2. Fourier coefficient component.
3. Convolution.
4. Fourier transform.

Exam skill

Recognizing the object is often half of the solution.

Integrated exercise: L^2 and Fourier coefficient

Exercise: Computation ***

Let $f(x) = x$ on $[-1, 1]$.

1. Compute $\|f\|_2$.
2. Check whether f is orthogonal to the constant function 1.

Solution: L^2 and orthogonality

Proof idea: Computation

$$\|f\|_2^2 = \int_{-1}^{1} x^2 dx = \frac{2}{3},$$

so

$$\|f\|_2 = \sqrt{\frac{2}{3}}.$$

Moreover

$$\langle f, 1 \rangle = \int_{-1}^{1} x \, dx = 0.$$

Therefore f is orthogonal to 1.

Common mistakes in Lesson 8

Counterexample: Mistake 1

Treating L^p elements as pointwise functions and forgetting equality almost everywhere.

Counterexample: Mistake 2

Using the L^2 inner product formula in L^1 or L^∞.

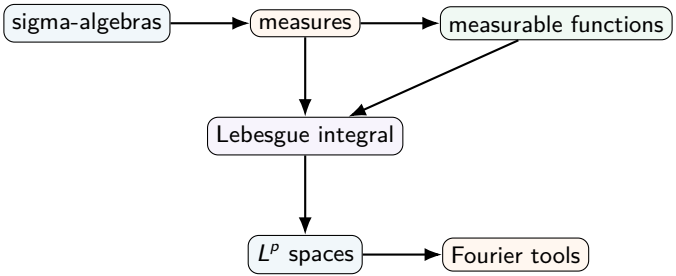
Counterexample: Mistake 3

Forgetting normalization constants in Fourier coefficients.

Counterexample: Mistake 4

Confusing Fourier series for periodic functions with Fourier transform on the real line.

Module 2 final map



Message

Module 2 turns integration into a geometric and analytical tool for functions.

End of Lesson 8: what to remember

Checklist

1. L^p spaces collect functions with finite p-th integral magnitude.
2. L^p elements are equivalence classes modulo equality almost everywhere.
3. Hölder controls products; Minkowski gives the triangle inequality.
4. L^2 is a Hilbert space with inner product $\int fg$.
5. Fourier coefficients are projections onto sine/cosine directions.
6. Plancherel preserves L^2 size.
7. Convolution becomes multiplication under the Fourier transform.